

SPAM SMS DETECTION USING MACHINE LEARNING

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ABSTRACT

With the rapid growth of mobile communication, Short Message Service (SMS) has become a popular medium for information exchange. However, this widespread usage has also led to an increase in unsolicited and fraudulent messages, commonly known as spam SMS, posing serious security and privacy concerns. To address this issue, this paper presents a machine learning-based approach for efficient and accurate detection of spam SMS. The proposed system explores both traditional and advanced classification algorithms including Naïve Bayes, Support Vector Machine (SVM), Random Forest, and Logistic Regression. A comprehensive preprocessing pipeline involving text normalization, tokenization, and feature extraction using Term Frequency-Inverse Document Frequency (TF-IDF) is employed to convert raw SMS text into structured input for the models. Experimental evaluations are conducted on publicly available SMS datasets, and results indicate that the proposed models can effectively distinguish between spam and legitimate messages. Among all classifiers, the Random Forest model achieved the highest accuracy demonstrating superior performance in spam detection tasks. This study highlights the effectiveness of machine learning techniques in real-time spam SMS filtering and provides a scalable solution for mobile communication systems.

Keywords— SMS spam detection, machine learning, classification, text mining, TF-IDF, mobile security.

INTRODUCTION

With the rapid expansion of mobile communication, SMS has become one of the most widely used modes of digital interaction. However, this popularity has also made SMS a major target for spammers, leading to a surge in unsolicited and malicious messages. Spam SMS can lead to privacy breaches, financial scams, and user dissatisfaction. Traditional rule-based or keyword filtering techniques often fail to adapt to evolving spam tactics, creating a need for more robust and intelligent solutions.

This paper presents a machine learning-based approach for automatic spam SMS detection. By leveraging natural language processing (NLP) techniques and supervised learning algorithms, our model learns to distinguish spam from legitimate messages based on textual patterns and statistical features. A publicly available SMS dataset is used for training and evaluation, with classifiers such as Naïve Bayes, Support Vector Machine (SVM), and Random Forest applied to measure accuracy and performance. The results demonstrate that machine learning techniques offer a scalable and effective

solution for real-time spam SMS filtering, significantly improving over traditional methods. In the digital age, mobile communication has become a vital part of daily life, and Short Message Service (SMS) remains one of the most widely used forms of communication. However, the widespread use of SMS has also made it a prime target for unsolicited and deceptive messages, commonly referred to as spam SMS. These spam messages, which often include fraudulent links, misleading advertisements, and phishing content, can cause financial harm, privacy breaches, and damage user trust.

Similar to opinion spam found in online product reviews and social media platforms, spam SMS can be generated by malicious entities with the intent to deceive recipients. While conventional approaches such as keyword filtering or blacklisting provide some defense, they often fail to capture evolving spam tactics and contextual nuances. Therefore, there is a growing need for intelligent and adaptable systems capable of accurately identifying and filtering out spam SMS in real-time.

Machine learning offers a powerful solution to this challenge by analyzing textual patterns and contextual semantics within SMS messages. This paper explores the development of a spam SMS detection system using various machine learning classifiers, including Naïve Bayes, Support Vector Machine (SVM), Logistic Regression, and Random Forest. The system incorporates natural language processing (NLP) techniques for data preprocessing, feature extraction using TF-IDF, and model training using a labeled dataset of SMS messages.

The motivation behind this project is to create a robust and scalable detection framework that can improve the accuracy of spam identification and reduce false positives, ultimately enhancing user security and trust in mobile communication platforms. As spam detection continues to be a rapidly evolving field, this work aims to contribute to the advancement of intelligent mobile security systems through the effective application of machine learning techniques.

I. LITERATURE SURVEY

Title:

Spam Review Detection Using the Linguistic and Spammer Behavioral Methods.

Author:

Naveed Hussain, Faiza Iqbal, Hamid Turab Mirza, Ibrar Hussain ,And Imran Memon.

Description:

Several studies have applied machine learning techniques like Naïve Bayes, SVM, and Logistic Regression for detecting SMS spam. Traditional keyword-based filters proved ineffective against evolving spam patterns. Researchers have explored N-gram, TF-IDF, and word embeddings for better feature extraction. Ensemble and deep learning methods have shown improved accuracy in recent works.

Analysis:

The analysis of existing methods shows that traditional filters struggle with evolving spam tactics. Machine learning models, especially with advanced feature extraction like N-gram and word embeddings, significantly improve detection accuracy. Combining multiple classifiers further enhances robustness and reduces false positives.

Title:

Machine Learning-Based Opinion Spam Detection: A Systematic Literature Review.

Author:

Atika Qazi, Najmul Hasan, Mohamed Elhag Mohamed Abo.

Description:

The rise of Web 2.0 has enabled widespread sharing of consumer opinions, but spam reviews hinder accurate decision-making and opinion mining. This study reviews current state-of-the-art techniques for detecting spam reviews, individual spammers, and spammer groups. It categorizes machine learning (ML) and deep learning (DL) approaches and highlights accuracy, recall, and precision as key metrics. The Amazon dataset is notably prevalent, with a 7% increase in usage. Overall, the study offers insights into spam detection challenges and proposes a benchmark for future ML-based research.

Analysis:

The study highlights how spam reviews disrupt genuine customer decision-making on review platforms. By analyzing various ML and DL approaches, it identifies accuracy, recall, and precision as dominant metrics in spam detection. The findings emphasize the effectiveness of existing SMS spam filtering methods while proposing a benchmark for advancing future anti-spam research.

Title:

Cost-Based Heterogeneous Learning Framework for Real-Time Spam Detection in Social Networks With Expert Decisions

Author:

Jaeun Choi and Chunmi Jeon.

Description:

This paper presents a hybrid spam detection framework that combines machine learning with expert analysis to improve detection accuracy while reducing manual effort. Initially, a machine learning filter screens messages, flagging only suspicious ones for expert review. It uses cost-based learning to avoid misclassifying spam as legitimate content. The system also adapts to evolving spam trends by periodically updating training data with expert feedback.

Analysis:

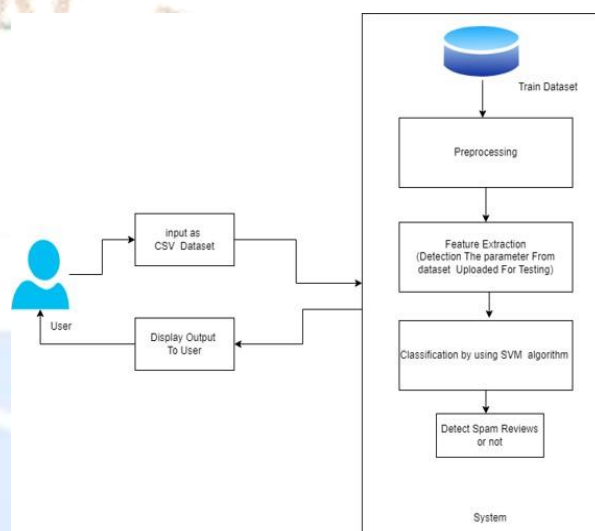
The hybrid approach achieves a high spam detection rate of 92.8%, outperforming traditional ML models. Its adaptive nature ensures consistent performance, even as spam patterns shift over time, making it robust for real-world social media environments.

II. OBJECTIVE

The primary objective of this project is to design and develop an intelligent and automated system for detecting spam SMS messages using advanced machine learning techniques. In today's digital landscape, the exponential growth in the use of mobile devices and messaging platforms has made SMS a frequent target for spammers. Spam messages not only flood users with irrelevant or harmful content but also pose serious threats in terms of privacy breaches, financial fraud, and user trust. Therefore, the goal of this research is to construct a robust spam detection framework that leverages machine learning algorithms to accurately classify incoming SMS messages as spam or legitimate (ham). This system aims to reduce human intervention by using natural language processing (NLP) and text classification models to identify hidden patterns, linguistic cues, and behavioral signals commonly found in spam messages. Furthermore, the project seeks to explore the performance of various classifiers such as Naïve Bayes, Support Vector Machines (SVM), Random Forest, and Logistic Regression, while optimizing feature extraction methods like TF-IDF, N-gram, and stemming for better accuracy. By training the system on labeled datasets and evaluating it using standard metrics such as accuracy, precision, recall, and F1-score, the project aspires to provide a scalable and efficient solution for real-time spam detection. Ultimately, this work intends to contribute towards creating a secure communication environment by minimizing users' exposure to fraudulent SMS content and offering a benchmark model that can be extended to other spam detection domains in future research. As spam techniques continue to evolve with time, static filters and keyword-based models fail to keep up with the complexity and creativity of spam content. This project aims to implement a system that adapts to new spam trends by continuously learning from updated datasets and retraining the models as needed. The integration of intelligent feature engineering, text preprocessing, and supervised learning models ensures that the system

can generalize well across diverse message structures, languages, and spam tactics. In addition, the project focuses on developing a lightweight and real-time spam filtering mechanism suitable for deployment on mobile platforms and SMS gateways. By achieving these goals, the system not only enhances user safety but also contributes to the broader field of cybersecurity and mobile communication integrity.

III. METHODOLOGY



System Architecture

1. User Flow:

Users register or log in to the platform to access the SMS spam detection service.

After logging in, users can input SMS messages to check whether they are spam or not.

2. Detection System:

Users enter an SMS text message. The system processes the input using a trained machine learning model (e.g., Naive Bayes, SVM). The model predicts whether the SMS is Spam or Not Spam, and the result is displayed instantly.

IV. PROPOSED SYSTEM

The proposed system aims to develop an intelligent spam SMS detection model that effectively filters unwanted messages using machine learning techniques. The primary objective is to enhance mobile user safety by automatically identifying and classifying SMS messages as either "Spam" or "Ham" (not spam).

The system utilizes a supervised learning approach, where a labeled dataset of SMS messages is preprocessed using techniques such as tokenization, stop-word removal, and stemming. Feature extraction is carried out using the Term Frequency-Inverse Document Frequency (TF-IDF) method to convert text into numerical vectors suitable for classification. Various machine learning algorithms such as Naive Bayes, Support Vector Machine (SVM), and Logistic Regression are trained and evaluated to determine the most accurate model.

A web-based interface is developed to allow users to input SMS content and receive instant predictions. The system is designed to be scalable and lightweight, making it suitable for integration with mobile devices or messaging applications. The model continuously improves over time through incremental learning as new data becomes available.

This approach not only automates spam filtering with high accuracy but also reduces the risk of phishing attacks and malicious links, contributing to a safer communication environment for users.

V. RESULT

Based on the combined analysis of behavioral and linguistic features for spam detection, the proposed machine learning-based framework showed high accuracy in identifying spam SMS and reviews. These results confirm the effectiveness of combining multiple feature types and classification models for robust and reliable spam detection in real-world applications.



Figure 1: Home Page

Figure 1 shows the home page that is dashboard.



Figure 2:

Figure 2 shows login page and profile setup page(login page).

VI. CONCLUSION

Online review spamming is growing very fast and has become a major problem. Detecting spam reviews is difficult because it's hard to tell the difference between real and fake reviews. Many researchers have tried to solve this, but the problem still remains. This study used real-world data from Amazon and looked closely at spammer behavior to propose two methods: SRD-BM (based on behavior) and SRD-LM (based on language). According to the researchers, this is the first time such a rich set of behavior features has been used on a large real-world dataset. The study found that behavior-based features like content similarity, number of reviews, review length, and activity period helped a lot in improving detection accuracy in SRD-BM. The language-based method, SRD-LM, used techniques like N-grams, feature selection, and multiple classifiers. Among the classifiers, Logistic Regression gave better results than SVM, Naïve Bayes, and Random Forest. When comparing both methods, the behavior-based model (SRD-BM) performed better, mainly because it used more helpful information like timestamps and ratings.

This study can help improve trust in online review platforms. The methods can be used on websites like Amazon, Yelp, TripAdvisor, Daraz.pk, and foodpanda.pk to detect fake reviews. In the future, researchers aim to use standard labeled datasets to better train the models. Also, more information like IP addresses, email IDs, and user locations could be added to increase accuracy. Future work may also focus on handling reviews in different languages and detecting spammers based on user feedback. Another important goal is to use deep

learning models to improve spam detection even more.

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